

Bernardo Justiça Godinho

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 Porto, Portugal



About me

Biomedical Engineer with experience in Python, data analysis and machine learning research. Experienced in developing and evaluating models using complex physiological datasets, with a strong interest in applying these skills to practical problems across different industries and developing reliable ML systems.

Experience

Researcher | INESC TEC

Contactless Sensing System for Health and Well-being Monitoring

Designed and tested a battery-powered capacitive ECG acquisition system, combining analog front-end and PCB design, signal acquisition, filtering and signal-quality analysis. Developed and experimentally validated the system from hardware prototype to physiological signal processing.

Apr 2026 - Sep 2026

ERASMUS+ Internship | Technical University of Denmark

Analysis of Heartbeat Evoked Potentials in Temporal Lobe Epilepsy

Processed and analysed ECG/EEG recordings using digital signal processing techniques, including filtering, time-frequency analysis and feature extraction in Python. Applied machine-learning models to physiological signals and evaluated their performance using cross-validation and explainability methods.

Sep 2025 - Feb 2026

Curricular Internship | INESC TEC

Computer Assisted Gastric Cancer Diagnosis Project

Collaborated in the implementation of a gastric cancer diagnosis model into hospital environments

Feb 2024 - Jun 2024

Academic Education

Master's Degree in Bioengineering - Biomedical Engineering

Thesis: "Contactless Sensing System for health and well-being monitoring"

Faculty of Engineering of Porto University

Sep 2024 - Jul 2026

Bachelor's Degree in Bioengineering

Faculty of Engineering of Porto University

Sep 2021 - Jul 2024

Publications

Godinho, B. & Correia, M.V., "A Single-Limb, Fully Battery-Powered Capacitive ECG Acquisition System" IEEE-EMBS BSN, 2026 (1st author)

Saibene, M., Godinho, B., et al., "Analysis of Heartbeat Evoked Potentials in Temporal Lobe Epilepsy," IEEE EMBC, 2026 (2nd author)

Complementary Education

XIII, XIV and XV Symposium in Bioengineering | NEB

2022-24

National Conference in Bioengineering | Engineering

Order - Northern Region (Porto) | ANEEB

2022

Junior University | Porto University

2013-2018

5th Degree in Piano | Academia de Espinho/Fórum Cultural de Gulpilhares

2013-2018

Languages

- Portuguese - Native
- English - C1 Level
 - Cambridge English Exam Advanced (CAE) - 2022

Technical Skills

Programming Languages

- Python, MATLAB, Java, C/C++

Machine Learning & Data Science

- Feature engineering, cross-validation, model evaluation, SHAP
- NumPy, pandas, scikit-learn, PyTorch

Data Analysis & Databases

- SQL, Matplotlib, seaborn

App & Web Development

- Android Studio, Firebase, Java
- PHP, HTML/CSS, JavaScript

Biosignal Analysis

- ECG, EEG, EMG
- MNE-Python, NeuroKit2, scikit-learn, SciPy, PyQt5

Development Tools

- Git, Docker, Flask, Firebase

Associativism

VO.U Volunteer

- VO.U for animals
- VO.U for nature

2022-2023

Student's Association

Escola Dr. Joaquim Gomes Ferreira Alves

2020-2021

Extracurricular Activities

Federated Tennis

2nd place, singles (2022) and doubles (2021), regional tournaments

2012-2025

University Basketball

ICBAS | FAP

2024-2026

Piano Jazz Lessons

2019-2022